

Aryan Dhillon

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EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Computer Engineering, Minor in Business Economics

Expected May 2027 / GPA: 3.51/4.00

Concentrations: Software Engineering, AI/ML

Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Object-Oriented Programming, Data Science, C/C++

WORK EXPERIENCE

Garmin

Olathe, KS

Software Engineering Intern

May 2026 – August 2026

- Resolved 8 formal change requests under Garmin's DO-178B airborne software certification process for GTS 8x0 v5.03, targeted for 2026 release, covering requirements, embedded C implementation, test updates, and independent review
- Contributed to TCAS mode synchronization, ADS-B In status validation, and traffic system self-test improvements across multiple libraries and codebases
- Updated the X-Plane traffic simulation plugin to support new aircraft simulation models, validating surveillance behavior through generated flight tracks

Vaila - AI Scheduling App | vaila.dev

March 2026 – Present

Founder & CEO

- Founded and independently built Vaila, an AI scheduling platform for iOS, using SwiftUI, React, FastAPI, PostgreSQL, and Redis, and prepared it for App Store launch.
- Designed a hybrid scheduling engine combining deterministic calendar conflict detection with OpenAI-powered ranking for 1:1, guest, and group scheduling
- Integrated Google Calendar, Outlook/Microsoft Graph, and Apple Calendar with OAuth, token refresh, availability syncing, provider entitlements, and confirmed event creation
- Architected production infrastructure with Redis-backed workers, PostgreSQL indexing, database migrations, connection-pool tuning, job tracking, and Sentry observability

The Data Mine (John Deere Project)

West Lafayette, IN

Undergraduate Data Science Researcher

August 2025 – May 2026

- Developed an LSTM-based multivariate time-series forecasting model to predict monthly replacement-part demand across 1,000+ part-location time series using 12-month historical demand sequences
- Engineered input features from historical demand and external economic indicators, including Equipment Made, Manufacturers' Equipment Inventory, and PPI Manufacturing, to improve forecast accuracy
- Evaluated probabilistic forecasts using RMSE, Mean Winkler Interval Score (MWIS), and 90% prediction interval coverage, and presented results at the Data Mine Corporate Partners Symposium

BEAMRail Solutions Group LLC (Startup)

Remote

Innovation Intern

April 2025 – August 2025

- Developed backend application components and automation pipelines for ID scan processing and report workflows, reducing officer reporting time significantly
- Redesigned dashboard UI/UX to streamline report submission, audio logging, and data ingestion workflows

PROJECTS

Distributed Key-Value Store (Java)

February 2026

- Built a distributed key-value store in Java with a custom open-addressing hash table (1.37 measured probes/lookup) and consistent-hash sharding (160 vnodes/node) for cluster-wide key distribution with minimal remapping on node changes
- Implemented crash-safe persistence via write-ahead logging with group commit, recovering 1.9M WAL records/sec on restart, and a non-blocking concurrent TCP server (java.nio) implementing the RESP protocol
- Engineered a lock-free, zero-allocation metrics pipeline (LongAdder counters, fixed-memory latency histograms) exposing p50/p99/p99.9 latency; instrumented a server sustaining 65K ops/sec across 32 concurrent clients

Investment Analytics Platform (Python)

September 2025

- Built an investment analytics and backtesting platform with a Streamlit frontend to simulate trading strategies on 5 years of historical OHLCV data from Yahoo Finance
- Designed a modular backtesting engine supporting 3 configurable strategies with portfolio state management, buy-and-hold benchmarking, Sharpe ratio, max drawdown, and return analysis
- Added risk controls including position sizing, stop-loss logic, regime filtering, and volatility filtering to evaluate strategy performance across different market conditions

SKILLS

Languages: C, C++, Python, Java, Swift, SQL | **Tools/Frameworks:** React, SwiftUI, FastAPI, Streamlit, Git, PostgreSQL, Redis, Pandas, NumPy, scikit-learn, TensorFlow/Keras, REST APIs, OAuth

